

CARDIAC ARREST (ACLS)

SUDDEN LOSS OF EFFECTIVE CARDIAC OUTPUT – SURVIVAL HINGES ON EARLY HIGH-QUALITY CPR AND EARLY DEFIBRILLATION, NOT THE DRUGS.

! WHAT IT IS

- Abrupt loss of mechanical cardiac function with no pulse or normal breathing, causing cessation of systemic and cerebral perfusion.
- Rhythms are either **shockable** (VF/pulseless VT) or **non-shockable** (asystole/PEA). Shockable arrests have the best survival.
- Irreversible brain injury begins within minutes.
- Most adult out-of-hospital arrests are cardiac (ischemic) in origin. In-hospital arrests and PEA are often due to reversible causes (the **Hs and Ts**).



SURVIVAL HINGES ON TWO THINGS



1 HIGH-QUALITY CPR

Push hard.
Push fast.
Minimize interruptions.



2 EARLY DEFIBRILLATION

Shock early for shockable rhythms.

Everything else is secondary.

THE CHAIN OF SURVIVAL



1 Early Recognition & Call for Help



2 Early High-Quality CPR



3 Early Defibrillation



4 Advanced Resuscitation



5 Post-Arrest Care



Early CPR and early defibrillation save the most lives.



PRIMARY GOALS

- ✓ Restore perfusion and oxygen delivery
- ✓ Identify and treat reversible causes (**Hs and Ts**)
- ✓ Achieve Return of Spontaneous Circulation (ROSC)
- ✓ Optimize post-arrest care to improve neurologic outcome



KEY FACTS

- CPR rate: **100–120/min**
- Compression depth: **at least 2 in (5 cm)** – avoid >6 cm
- Allow full chest recoil
- Rotate compressor every **2 min** (at rhythm check)
- Epinephrine every **3–5 min**
- Minimize no-flow time



OUTCOMES THAT MATTER

- ROSC (short-term goal)
- Survival to hospital discharge
- Good neurologic outcome (CPC 1–2)



Quality of CPR and early shock move the needle.



RED FLAG

Agonal gasping is **NOT** breathing. Treat as cardiac arrest and start CPR immediately.



ETCO₂ is your real-time window.

Persistent ETCO₂ <10 mmHg despite optimized CPR is associated with poor likelihood of ROSC. Sudden rise = possible ROSC.

01

IMAGE TITLE
Cardiac Arrest (ACLS) Overview

CATEGORY
Cardiology

DIFFICULTY
Foundation

CLINICAL SOURCE
AHA Guidelines for CPR & ECC (2025)
Adult Advanced Life Support

REVIEW
CLINICAL PEARLS

CARDIAC ARREST (ACLS)

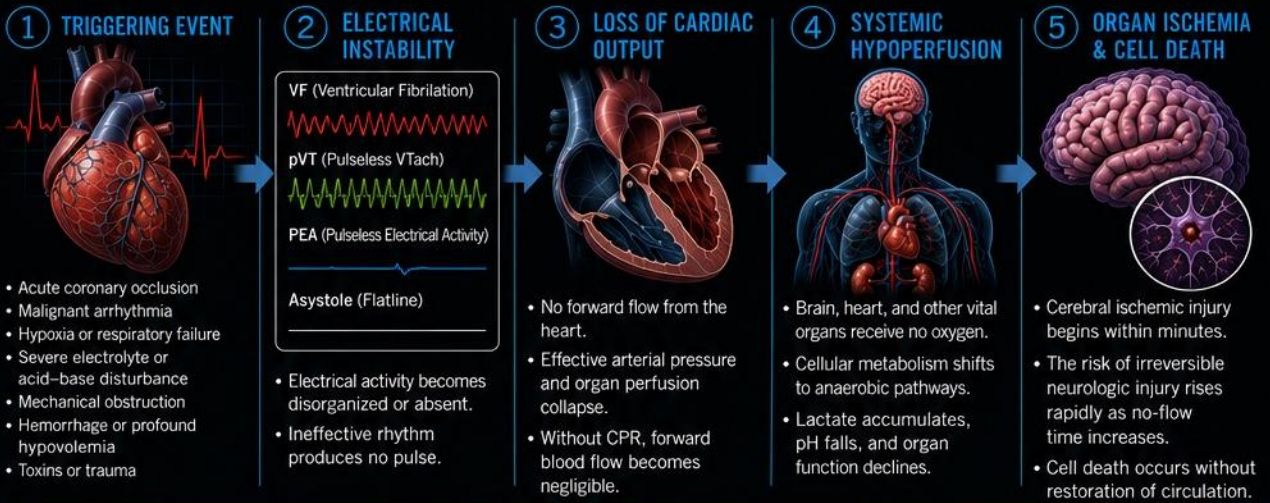
PATHOPHYSIOLOGY: FROM TRIGGER TO GLOBAL ISCHEMIA

An acute event triggers electrical instability, leading to loss of effective pumping, systemic hypoperfusion, and progressive organ ischemia.



KEY CONCEPT

Cardiac arrest is not a single event but a rapid cascade. Immediate, high-quality CPR can interrupt the downward spiral and preserve life and brain function.



PERFUSION DURING CPR: THE GOAL IS OXYGEN TO THE BRAIN AND HEART

CEREBRAL PERFUSION

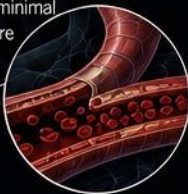
Chest compressions generate pulsatile blood flow to the brain through carotid and vertebral arteries during decompression.

Internal carotid artery

Vertebral artery

CORONARY PERFUSION

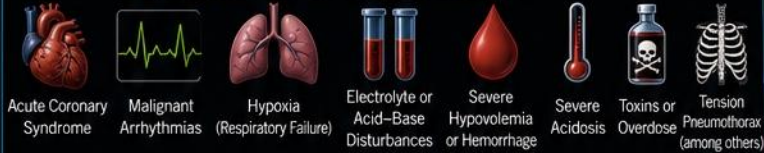
Coronary blood flow occurs mostly during chest recoil (decompression) when aortic pressure exceeds right-atrial pressure. Full recoil and minimal interruptions are essential.



CPR PHYSIOLOGY SUMMARY

- 
Compressions act as an external heart, generating artificial circulation.
 - 
Full recoil promotes venous return and helps restore the pressure gradient required for coronary and cerebral perfusion.
 - 
Quality CPR preserves brain and heart tissue until ROSC is achieved.
 - 
Goal: Maintain perfusion to vital organs and buy time for definitive care.

COMMON CAUSES OF CARDIAC ARREST



REVERSIBLE CAUSES: H's & T's

- H's

 - Hypovolemia
 - Hypoxia
 - Hydrogen ion (Acidosis)
 - Hyper/ Hypokalemia
 - Hypothermia

T's

 - Tension Pneumothorax
 - Tamponade, cardiac
 - Toxins
 - Thrombosis, pulmonary
 - Thrombosis, coronary

THE VICIOUS CYCLE OF CARDIAC ARREST



HOW CPR INTERRUPTS THE CYCLE





RED FLAGS


Unresponsive patient


No normal breathing or only agonal gasps


No definite central pulse within 10 seconds


Every second counts. Start CPR immediately.

CARDIAC ARREST (ACLS)

CLINICAL PRESENTATION: RECOGNITION IS CRITICAL

Cardiac arrest often presents as sudden collapse but may also follow progressive respiratory or hemodynamic deterioration.

Early recognition and immediate action are the keys to survival.

CLASSIC SUDDEN COLLAPSE



Abrupt loss of consciousness



Collapse



No response to tapping shoulders and shouting



No normal breathing or only gasping (agonal respirations)



No definite pulse within 10 seconds

CHECK RESPONSE

Tap shoulder and shout.

CHECK BREATHING & PULSE

Look for no normal breathing or only gasping while checking carotid pulse simultaneously for no more than 10 seconds.

NO NORMAL BREATHING & NO PULSE

Start CPR immediately.



KEY CONCEPT

Cardiac arrest is a clinical diagnosis based on unresponsiveness, absence of normal breathing, and no definite pulse.

Do not delay — start CPR.

AGONAL RESPIRATIONS

Occasional gasps or gurgling sounds are not normal breathing and are a sign of cardiac arrest.



DO NOT MISS CARDIAC ARREST

- ✗ Agonal gasps are not normal breathing.
- ✗ Brief seizure-like movements may occur at arrest onset.
- ✗ Do not assume a pulse based on movement or gasping.
- ✗ When uncertain, begin CPR.

OTHER SIGNS YOU MAY SEE

PALE OR CYANOTIC SKIN



Poor perfusion leads to pallor, mottling, or a bluish color.

FIXED, DILATED PUPILS



Late, nonspecific finding. May develop during prolonged no-flow states. Do not delay CPR or guide early prognosis.

LIMP OR FLACCID BODY



Loss of muscle tone and no spontaneous movement.

AGONAL GASPS



Irregular, gasping breaths may occur early and must not be mistaken for normal breathing.

ABSENT PULSE



No definite carotid pulse felt within 10 seconds.

MONITOR FINDINGS



These rhythms confirm cardiac arrest. Do not delay CPR while analyzing.

RAPID ASSESSMENT ALGORITHM (HEALTHCARE PROVIDER)



1

SCENE SAFETY
Ensure the area is safe for you and the victim.



2

CHECK RESPONSE
Tap shoulder and shout. No response?



3

ACTIVATE & SEND FOR AED
Activate emergency response system. Send another responder for AED.



4

CHECK BREATHING & PULSE
Look for no normal breathing or only gasping while checking carotid pulse simultaneously for ≤10 seconds. No normal breathing and no pulse?



5

START CPR
Begin high-quality CPR immediately. Use AED as soon as it arrives.

IMMEDIATE ACTIONS (DO ALL WITHOUT DELAY)



1

Start high-quality CPR immediately.



2

Activate the emergency response system (call for help).



3

Send another responder for the AED/defibrillator.



4

Attach monitor/defibrillator when available and follow the ACLS algorithm.

HIGH-VIGILANCE SITUATIONS: THE PATIENT MAY BE DETERIORATING



Unresponsive or confused



Progressive hypoxemia



Severe hypotension



Marked bradycardia



Ventricular arrhythmia



Sudden unresponsiveness



Peri-arrest deterioration



Post-cardiac surgery



High-risk cardiac disease



RED FLAGS



Unresponsive patient



No normal breathing or only agonal gasps



No definite central pulse within 10 seconds



Every second counts. Start CPR immediately.

04

IMAGE TITLE

Clinical Presentation: Recognition is Critical

CATEGORY

Cardiology

DIFFICULTY

Foundation

CLINICAL SOURCE

AHA Guidelines for CPR & ECC (2025)
Adult Advanced Life Support

REVIEW

CLINICAL
PENDING

CARDIAC ARREST (ACLS)

RAPID ARREST ASSESSMENT & REVERSIBLE-CAUSE SEARCH

Confirm arrest within 10 seconds, begin CPR, and search for reversible causes without interrupting compressions.

1
CONFIRM CARDIAC ARREST (≤ 10 SECONDS)

RESPONSIVENESS

No response to tapping shoulders and shouting.

BREATHING

No normal breathing; only agonal gasps or absent.

PULSE (SIMULTANEOUS)

Check carotid pulse while assessing breathing for no more than 10 seconds.

NO NORMAL BREATHING AND NO DEFINITE PULSE: START CPR IMMEDIATELY.

Check responsiveness

Tap shoulders and shout.

Check breathing

Look for no normal breathing or only agonal gasps.

Check carotid pulse

Simultaneously for ≤ 10 seconds. No definite pulse.

KEY CONCEPT

Cardiac arrest is recognized clinically by unresponsiveness, absent normal breathing, and no definite pulse. Assess rapidly (≤ 10 seconds), start CPR, and search for reversible causes without delay.

2
QUICK FOCUSED EXAM

- GENERAL APPEARANCE

Pale, cyanotic, or mottled skin.

SKIN

Cool, clammy, or cyanotic skin.

NECK

JVD may suggest obstructive physiology (e.g., tamponade, tension pneumothorax), but absence of JVD does not exclude it.

PUPILS

Pupillary findings do not diagnose arrest or determine early prognosis.

LUNGS

Equal chest rise (if any). Listen only if it does not delay CPR.

HEART

Auscultation is not required and should not delay CPR.

PHYSICAL EXAM MUST NOT DELAY CPR.

3
SEARCH FOR REVERSIBLE CAUSES (H's & T's)
SEARCH WHILE CPR CONTINUES — DO NOT PAUSE COMPRESSIONS.

H's HYPOVOLEMIA Look for signs of blood loss or dehydration.	HYPOXIA Check airway patency, ventilation, oxygen source & delivery, tube position, and waveform capnography (if advanced airway).	HYDROGEN ION (ACIDOSIS) Consider prolonged arrest, shock, or respiratory failure.	HYPO-/HYPERKALEMIA Review history, labs (if available), and medications.	HYPOTHERMIA Check core temperature; look for cold environment.
T's TENSION PNEUMOTHORAX Sudden ventilation difficulty, unilateral absent breath sounds, asymmetric chest movement, high airway pressure, or compatible US findings. Treat immediately if suspected.	CARDIAC TAMPONADE Consider trauma, recent procedure, malignancy, or pericardial disease. Use focused US (if available) without interrupting CPR.	TOXINS Consider overdose, poisoning, or medication effects; review history, look for clues.	THROMBOSIS (PULMONARY) Sudden dyspnea, hypoxia, risk factors for PE.	THROMBOSIS (CORONARY) History of CAD, chest pain, risk factors.

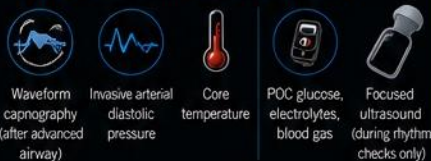
4
MONITOR & OBJECTIVE DATA

ATTACH MONITOR / DEFIBRILLATOR



Identify rhythm, but do not delay CPR while attaching.

PHYSIOLOGIC DATA — WHEN AVAILABLE



Never prolong a compression pause to obtain data.

EXAM PEARLS

- Speed saves lives—complete the initial assessment in ≤ 10 seconds.

Begin CPR if there is no normal breathing and no definite pulse.

Do not allow the search for a pulse to exceed 10 seconds.

Auscultation of heart or breath sounds must not delay CPR.

Use the H's & T's to rapidly consider reversible causes.

Reassess rhythm and response approximately every 2 minutes.

COMMON PITFALLS

- Spending too much time checking pulse or looking for signs.

Waiting for monitoring before starting CPR.

Mistaking agonal gasps for normal breathing.

Delaying CPR while searching for reversible causes.

Not considering H's & T's early.

Interrupting compressions for non-essential tasks.

RED FLAGS:

Unresponsive patient

No normal breathing or only agonal gasps

No definite central pulse within 10 seconds

Every second counts. Start CPR now.

05	IMAGE TITLE Physical Exam: Rapid Assessment & Priorities	CATEGORY Cardiology	DIFFICULTY Foundation	CLINICAL SOURCE AHA Guidelines for CPR & ECC (2025) Adult Advanced Life Support	REVIEW CLINICAL PENDING
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CARDIAC ARREST (ACLS)

RHYTHM DIAGNOSIS, PHYSIOLOGIC MONITORING & REVERSIBLE-CAUSE TESTING

Testing is targeted and subordinate to uninterrupted CPR, rapid rhythm classification, defibrillation, and reversible-cause treatment.

1 ACTIVE ARREST ASSESSMENT (OBTAIN WITHOUT INTERRUPTING CPR)

MONITOR / DEFIBRILLATOR

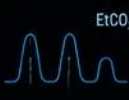


Real-time rhythm analysis and defibrillation.

Confirm asystole in second lead.

Check leads, cables, and gain.

WAVEFORM CAPNOGRAPHY



Confirms and monitors advanced airway, reflects CPR quality and perfusion.

Abrupt rise may indicate ROSC.

POINT-OF-CARE ULTRASOUND (POCUS)



- Pericardial effusion (tamponade physiology)
- RV dilation in context
- Lung sliding (tension pneumothorax)
- Volume status clues
- Cardiac motion may inform assessment (do not use alone to terminate resuscitation)

Acquire only during scheduled checks (≤ 10 sec).

INVASIVE ARTERIAL PRESSURE



Diastolic BP can help assess perfusion and CPR quality when available.

POINT-OF-CARE GLUCOSE

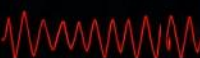


Treat severe hypo-/hyperglycemia promptly.

(Check early and repeat as needed.)

2 CARDIAC RHYTHM IDENTIFICATION

VF (VENTRICULAR FIBRILLATION)



Chaotic waveform with no identifiable QRS complexes. Shockable.

PULSLESS VT (PVT)



Regular or polymorphic wide-complex tachycardia with no pulse. Shockable.

PEA (PULSELESS ELECTRICAL ACTIVITY)



Organized electrical rhythm (P waves or QRS) with no palpable pulse. Not shockable.

ASYSTOLE



Near-flat line, no electrical activity. Not shockable.

PEA is a clinical diagnosis: organized electrical activity with no palpable pulse.

Confirm asystole in another lead and check monitor gain and connections.

4 TARGETED LABORATORY TESTS (OBTAIN EARLY WHEN AVAILABLE)

BLOOD GAS (ABG / VBG)



Assess pH, PaCO₂, PaO₂, HCO₃⁻, lactate. Guide ventilation and perfusion.

ELECTROLYTES (K⁺, Na⁺, Ca²⁺, Mg²⁺, Cl⁻)



Identify and treat life-threatening abnormalities.

GLUCOSE



Hypoglycemia can cause or worsen arrest. Treat promptly.

CBC



Identify anemia, infection, or bleeding that may contribute.

LACTATE



Marker of hypoperfusion. Trend to follow response.

TROPONIN



Often elevated after arrest. Interpret only after ROSC.

TOXICOLOGY



Consider overdose, poisoning, or exposure in appropriate scenarios.

Obtain these tests in parallel without interrupting CPR.

6 IMAGING (PERFORM AFTER ROSC OR WHEN STABLE)

CHEST X-RAY



Assess lungs, heart size, tube/line position, pneumothorax, edema, or aspiration.

ECHOCARDIOGRAPHY



Cardiac function, wall motion, valvular disease, pericardial effusion.

CT PULMONARY ANGIOGRAPHY



Use to evaluate suspected pulmonary embolism when stable.

NONCONTRAST HEAD CT



Evaluate suspected intracranial hemorrhage or mass when stable.

OTHER IMAGING



Obtain based on clinical context after ROSC or during stabilization.

Imaging should not delay CPR or defibrillation.



KEY CONCEPT

Use a focused, rapid diagnostic approach. Obtain targeted tests in parallel without interrupting compressions. Imaging and definitive testing are performed after ROSC or when the patient is stable.

3 REVERSIBLE-CAUSE SEARCH (Hs & Ts)

H's



HYPOVOLEMIA

Look for signs of blood loss or dehydration.



HYPOXIA

Assess airway patency, oxygen source, ventilation, and advanced-airway position. Use capnography.



HYDROGEN ION (ACIDOSIS)

Consider prolonged arrest, shock, or respiratory failure.



HYPO-/HYPERKALEMIA

Check electrolytes when available. Treat abnormalities.



HYPOTHERMIA

Measure core temp. Severe hypothermia may cause arrest. Use modified resuscitation and active rewarming.

T's



TENSION PNEUMOTHORAX

Sudden ventilation difficulty, unilateral absent breath sounds, high airway pressure. Treat immediately.



CARDIAC TAMPONADE

Consider trauma, recent procedure, malignancy, or pericardial disease. Use focused ultrasound when available.



TOXINS

Consider overdose, poisoning, or medication effects. Review history. Look for clues.



THROMBOSIS (PULMONARY)

Sudden dyspnea, hypoxia, risk factors. Treat if strongly suspected.

5 PHYSIOLOGIC MONITORING & CONFIRMATION DATA



WAVEFORM CAPNOGRAPHY

Confirms/monitors airway, provides physiologic feedback. Abrupt sustained rise may indicate ROSC. Low values → reassess CPR quality and causes.



INVASIVE ARTERIAL PRESSURE

Diastolic BP > 20 mmHg suggests adequate perfusion during CPR (if available).



12-LEAD ECG (AFTER ROSC)

Identify STEMI/NSTEMI, arrhythmias, ischemia, or other abnormalities.



FOCUSED ULTRASOUND (AFTER ROSC)

Assess cardiac function, volume status, and ongoing reversible causes.



URINE OUTPUT (AFTER ROSC)

Monitor end-organ perfusion and response.

7 PILLARS FOR TESTING DURING ARREST



- ✓ Begin CPR immediately.
- ✓ Reassess and treat reversible causes.
- ✓ Minimize interruptions.
- ✓ Recheck rhythm ≈ every 2 minutes.
- ✓ Use point-of-care tools.
- ✓ Limit pauses to ≤ 10 seconds.



RED FLAGS:



Unresponsive patient



No normal breathing or only agonal gasps



No definite central pulse within 10 seconds



Every second counts. Start CPR now.

ADULT CARDIAC ARREST ALGORITHM

HIGH-QUALITY CPR • RAPID RHYTHM RECOGNITION • REVERSIBLE CAUSES • TEAMWORK

CPR QUALITY: Push hard (at least 2 inches / 5 cm) • Push fast (100–120/min) • Allow full recoil • Minimize pauses • Rotate compressors every 2 min



1

RECOGNIZE ARREST

Unresponsive • No normal breathing
No definite pulse (≤ 10 sec)



2

START CPR

High-quality CPR
Give oxygen • Attach monitor / defibrillator
Consider advanced airway • IV/IO access



3

RHYTHM CHECK

Check rhythm every ~2 minutes
Check pulse if organized rhythm or
signs of ROSC (≤ 10 sec pause)



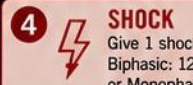
KEY PEARLS

- Begin CPR immediately.
- ✓ High-quality CPR is the priority.
- ✓ Check rhythm every ~2 minutes.
- ✓ Limit pauses to ≤ 10 seconds.
- ✓ Epinephrine every 3–5 minutes.
- ✓ Treat reversible causes.
- ✓ Capnography helps guide CPR quality and ROSC.
- ✓ Teamwork and communication save lives.

SHOCKABLE (VF / PULSELESS VT)

VF (Ventricular Fibrillation)

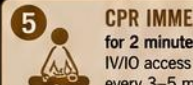
pVT (Pulseless VT)



4

SHOCK

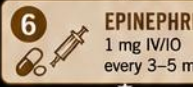
Give 1 shock
Biphasic: 120–200 J
or Monophasic: 360 J



5

CPR IMMEDIATELY

for 2 minutes
IV/IO access • Epinephrine
every 3–5 minutes



6

EPINEPHRINE

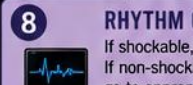
1 mg IV/IO
every 3–5 minutes



7

CONSIDER AMIODARONE

300 mg IV/IO bolus
Then 150 mg IV/IO bolus



8

RHYTHM CHECK

If shockable, go to Step 4.
If non-shockable,
go to appropriate branch.

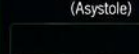
DURING CPR

- ✓ Ensure high-quality CPR
- ✓ Minimize interruptions
- ✓ Waveform capnography (ETCO₂ monitoring)
- ✓ Low ETCO₂ should prompt reassessment of CPR quality.
- ✓ Optimize oxygenation and ventilation
- ✓ IV/IO access
- ✓ Give epinephrine every 3–5 minutes
- ✓ Search for reversible causes (Hs & Ts)

NON-SHOCKABLE (PEA / ASYSTOLE)

PEA
(Pulseless Electrical Activity)

ASYSTOLE
(Asystole)



4

CPR IMMEDIATELY

for 2 minutes
IV/IO access • Epinephrine
every 3–5 minutes



5

RHYTHM CHECK

If still PEA/Asystole,
continue CPR and go to
Step 4. Check every 2 min.

REVERSIBLE CAUSES (Hs & Ts)

- Hs**
- Hypovolemia
 - Hypoxia
 - Hydrogen ion (acidosis)
 - Hypo-/hyperkalemia
 - Hypothermia
- Ts**
- Tension pneumothorax
 - Cardiac tamponade
 - Toxins
 - Thrombosis (pulmonary)
 - Thrombosis (coronary)

Treat causes
while continuing
CPR.

RETURN OF SPONTANEOUS CIRCULATION (ROSC)

Pulse and/or BP • Abrupt sustained rise in ETCO₂ • Spontaneous movement
MAP ≥ 65 mmHg

POST-CARDIAC ARREST CARE (AFTER ROSC)

Optimize
oxygenation and
ventilation
SpO₂ 94–88%
PaCO₂ 35–45 mmHg

Hemodynamic
stabilization
MAP ≥ 65 mmHg

Temperature
control
Prevent fever
32–37.5°C
per protocol

Glucose
control
140–180 mg/dL

12-lead ECG
and consider
coronary
angiography

Ongoing

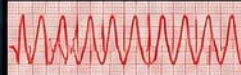
RHYTHM RECOGNITION GUIDE

VF (VENTRICULAR FIBRILLATION)

pVT (PULSELESS VT)

PEA (PULSELESS ELECTRICAL ACTIVITY)

ASYSTOLE



Chaotic, no identifiable
QRS complexes.

Wide-complex tachycardia,
no pulse.

Organized electrical activity
(without pulse).

Near-flat line, no electrical activity.
Confirm in another lead.



RED FLAGS

Unresponsive
patient

No normal
breathing or
only agonal gasps

No definite pulse
within 10 seconds

Every second counts.
Start CPR now.

07

IMAGE TITLE
Adult Cardiac Arrest
Algorithm

CATEGORY
Cardiology

DIFFICULTY
Foundation

CLINICAL SOURCE
AHA Guidelines for CPR & ECC (2025)
Adult Advanced Life Support

REVIEW
CLINICAL
PENDING

HISTOLOGY OF CARDIAC ARREST

GLOBAL HYPOXIC-ISCHEMIC & REPERFUSION INJURY

Cardiac arrest causes global hypoperfusion and cellular energy failure. Histologic changes vary by tissue, duration of ischemia, and reperfusion. Correlate pathology with clinical context, rhythm, and resuscitation data.



KEY PEARLS

- Histology cannot determine the exact time of arrest.
- Findings depend on cause, duration, temperature, reperfusion, and prior disease.
- Many changes are not appreciable until hours later.
- Always interpret in the clinical context.

① TIME COURSE OF MYOCARDIAL ISCHEMIC INJURY (APPROXIMATE)

MINUTES TO SEVERAL HOURS	~4-12 HOURS EARLY NECROSIS	~12-24 HOURS INCREASING NECROSIS	1-3 DAYS NEUTROPHILIC INFILTRATE	3-7 DAYS MACROPHAGE PHASE	1-2 WEEKS & BEYOND GRANULATION & SCAR
- Biochemical & ultrastructural injury - Routine H&E may be normal or nonspecific - Reversible if perfusion restored	- Loss of striations - Wavy fibers - Hyper-eosinophilic cytoplasm - Interstitial edema and hemorrhage	- Hyper-eosinophilic myocytes with pyknotic nuclei - Early neutrophils - Contraction bands may be seen	- Coagulative necrosis established - Dense neutrophilic infiltration - Hemorrhage may be present	- Macrophages remove necrotic tissue - Granulation tissue begins to form	- Granulation tissue develops - Collagen deposition - Mature fibrous scar over weeks-months

POTENTIALLY REVERSIBLE

IRREVERSIBLE

① Irreversible myocardial injury can begin after prolonged severe ischemia, but the timing is variable and cannot be assigned a universal cutoff.

② SUBENDOCARDIAL VULNERABILITY

The subendocardium is most vulnerable due to high wall stress and relatively reduced perfusion.

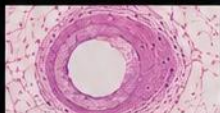
Global hypoperfusion may produce diffuse subendocardial injury, but patterns vary with cause and duration.



③ ONE POSSIBLE CAUSE: ACUTE CORONARY THROMBOSIS *

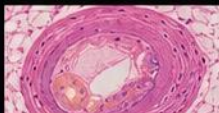
* Not all cardiac arrests are caused by coronary plaque rupture.

A. NORMAL CORONARY ARTERY



- Intact endothelium
- Thin intima
- Patent lumen
- No atherosclerosis

B. ATHEROSCLEROTIC PLAQUE



- Fibrous cap
- Lipid core
- Calcification may be present
- Narrowed lumen

C. PLAQUE RUPTURE & THROMBUS



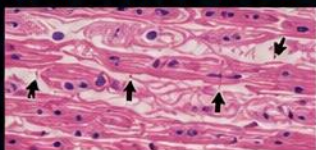
- Fibrous cap disruption
- Platelet-fibrin thrombus
- Occlusive luminal thrombus
- Underlying necrotic core

④ GLOBAL HYPOXIC-ISCHEMIC INJURY (MULTI-ORGAN)

BRAIN (NEURONS)	LUNGS (ALVEOLI)	LIVER (HEPATOCYTES)	KIDNEY (TUBULES)
- Eosinophilic neuronal injury (red neurons) - Shrinkage - Loss of Nissl substance - Occurs after a delay (hours)	- Interstitial edema - Intra-alveolar edema or hemorrhage - May reflect pulmonary edema, aspiration, or ARDS	- Centrilobular (zone 3) hypoxic injury - Ballooning degeneration - Spotty necrosis - May progress over hours to days	- Acute tubular injury - Loss of brush border - Tubular dilation - Cast formation - Findings evolve over time

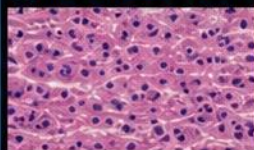
① Findings depend on duration and severity of hypoperfusion and may evolve over hours to days after ROSC.

⑤ CONTRACTION-BAND NECROSIS



- Hypercontraction injury associated with reperfusion or catecholamine excess
- Eosinophilic transverse bands (arrows)
- Supportive finding; not specific for mechanism of arrest
- May be seen early after insult

⑥ REPERFUSION INJURY



- Reperfusion can cause additional cellular damage
- Neutrophils, edema, and microvascular injury
- Oxidative stress and calcium overload contribute

⑦ ANCILLARY STAINS (SELECTED USE)



Ancillary stains may support interpretation in selected pathology or forensic cases but are not required for clinical diagnosis of cardiac arrest.

STAIN / TEST	PURPOSE / FINDING	LIMITATIONS
Masson Trichrome	Highlights fibrosis (collagen)	Not specific for timing
Periodic Acid-Schiff (PAS)	Glycogen depletion in ischemia	Variable
TUNEL Stain	Detects DNA fragmentation (apoptosis)	Not routine; may be negative early
CD68 (Macrophage marker)	Highlights macrophages in resorption phase	Appears days after injury
cTnl / cTnT IHC	Confirms myocardial origin in selective cases	Not required for routine evaluation

⑧ WHY HISTOLOGY CANNOT TIME THE ARREST

- Histologic changes have a wide time range and overlap.
- Many early changes are subtle or absent on routine H&E.
- Reperfusion, temperature, comorbidities, and sampling location affect appearance.
- Use multimodal information: rhythm, clinical history, downtime, biomarkers, imaging, and exam findings.



RED FLAGS



Prolonged no-flow time



Severe hypotension



Hypoxia (SpO₂ < 90%)



Severe acidosis (pH < 7.1)



Hyperthermia



Toxic exposure or overdose

TREATMENT OF CARDIAC ARREST

ACLS INTERVENTIONS & SUPPORTIVE THERAPY

High-quality CPR, early defibrillation, vasopressors, and post-ROSC care are the foundations of successful resuscitation.



KEY PEARLS

- High-quality CPR is the most important intervention.
- Minimize interruptions and provide early defibrillation.
- Epinephrine and amiodarone are lifesaving in arrest.
- Post-ROSC care improves survival and neurologic recovery.

IMMEDIATE PRIORITIES



High-quality CPR
Minimize interruptions
Rotate every 2 min



Early Defibrillation
As soon as possible
for shockable rhythms



Vascular Access
IV/IO access for
medications & fluids



Monitor & Reassess
Rhythm, ETCO₂, and
response to therapy

SHOCKABLE RHYTHMS (VF / PULSELESS VT)



- Shock**
Biphasic: 120–200 J
Monophasic: 360 J
- CPR for 2 minutes**
High-quality CPR
- Epinephrine** 1 mg IV/IO
every 3–5 minutes
- Amiodarone** 300 mg IV/IO
If still VF/pVT, give
150 mg IV/IO
- Continue CPR & rhythm
checks every ~2 minutes

NON-SHOCKABLE RHYTHMS (PEA / ASYSTOLE)



- CPR for 2 minutes**
High-quality CPR
- Epinephrine** 1 mg IV/IO
every 3–5 minutes
- Consider advanced airway,
capnography, and reversible
causes (Hs & Ts)
- Continue CPR & rhythm
checks every ~2 minutes



MEDICATIONS

Epinephrine (First-line)
1 mg IV/IO every 3–5 minutes

Amiodarone (Shockable)
300 mg IV/IO bolus
Then 150 mg IV/IO bolus

Lidocaine (Alternative)
1–1.5 mg/kg IV/IO bolus
Then 0.5–0.75 mg/kg

*Repeat or alternate antiarrhythmic
for refractory VF/pVT.*

SUPPORTIVE INTERVENTIONS

AIRWAY & OXYGENATION



- Provide 100% oxygen
during resuscitation.
- After ROSC, titrate
oxygen to maintain
SpO₂ 92–98% while
avoiding hyperoxia.
- Advanced airway
placement must not
interrupt chest
compressions.

MONITORING



- Continuous ECG
monitoring
- ETCO₂ goal ≥10 mmHg
(indicates effective
CPR)
- Pulse oximetry
- Blood pressure
(when ROSC achieved)

VASCULAR ACCESS



- IV or IO access as
soon as possible
- IO if IV access
cannot be obtained
quickly
- Use large-bore IV
when available

FLUIDS



- Isotonic crystalloid
250–500 mL bolus
for hypovolemia
- Avoid routine fluids
in absence of
hypovolemia
- Titrate based on
clinical picture

ADVANCED CONSIDERATIONS



- Ultrasound to identify
reversible causes
- Mechanical CPR device
if available
- Extracorporeal CPR
in selected cases
- Consult cardiac arrest
team or ECMO center

REVERSIBLE CAUSES (HS & TS)

Hs – HYPOXIA RELATED



Hypovolemia



Hypoxia



Hydrogen ion
(acidosis)



Hypo-/
Hyperkalemia



Hypothermia

Ts – OTHER CAUSES



Tension
pneumothorax



Cardiac
tamponade



Toxins



Thrombosis
(pulmonary)



Thrombosis
(coronary)

DEFIBRILLATION ENERGY (ADULT)

TYPE	BIPHASIC*	MONOPHASIC
Initial Shock	120–200 J	360 J
Second Shock	120–200 J	360 J
Subsequent Shocks	120–200 J	360 J

* Use the manufacturer's recommended biphasic energy when known; if unknown, use the maximum available setting.

CPR QUALITY TARGETS

 Rate 100–120 compressions per minute	 Depth At least 5 cm; avoid >6 cm	 Recoil Allow complete chest recoil	 Minimize Interruptions Pause <10 sec
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ETCO₂ GUIDANCE

- Persistently low ETCO₂ should prompt reassessment of CPR quality, ventilation, and reversible causes.
- A sudden, sustained increase in ETCO₂ may indicate ROSC.

POST-RETURN OF SPONTANEOUS CIRCULATION (ROSC) CARE

OPTIMIZE OXYGENATION AND VENTILATION



- SpO₂ 92–98%
- PaCO₂ 35–45 mmHg
- Avoid hyperoxia
and hypocapnia

HEMODYNAMIC SUPPORT



- Maintain MAP ≥65 mmHg
- IV fluids and/or vasopressors
as needed
- Titrate to perfusion goals

TEMPERATURE CONTROL



- Prevent fever
- Temperature control and
fever prevention
32–37.5°C according to
institutional protocol

TREAT CAUSE



- Identify and treat
the underlying cause
- Consider coronary
angiography if
indicated

NEUROLOGIC CARE



- Monitor neurologic status
- Avoid hyperglycemia
(140–180 mg/dL)
- Seizure management
if needed

ONGOING CARE



- ICU admission
- Continuous
monitoring
- Multidisciplinary
team care



AVOID / RED FLAGS



Prolonged
interruptions
in chest
compressions



Delayed or
missed
defibrillation



Hypoxia,
hyperoxia,
or hypocapnia



Failure to treat
reversible
causes



Premature
discontinuation
of efforts



Failure to
recognize
ROSC

CLINICAL PEARL

KEY TAKEAWAYS FOR SUCCESSFUL RESUSCITATION

Cardiac arrest survival depends on rapid recognition, high-quality CPR, early defibrillation, treatment of reversible causes, and excellent post-ROSC care. Prepare, perform, and persist—together as a team.



CLINICAL PEARLS

- High-quality CPR is the single most important factor.
- ✓ Every second without CPR reduces the chance of survival.
- ✓ Early defibrillation offers the greatest chance of survival in shockable rhythms.
- ✓ Treat reversible causes while performing excellent CPR.
- ✓ Post-ROSC care is critical for good neurologic outcome.
- ✓ Teamwork, communication, and leadership save lives.

THE CHAIN OF SURVIVAL



1. RECOGNITION

Early recognition and activation of emergency response.

2. HIGH-QUALITY CPR

Push hard, push fast, minimize interruptions, allow full recoil.

3. EARLY DEFIBRILLATION

Defibrillate as soon as possible for shockable rhythms.

4. ADVANCED RESUSCITATION

ACLS, medications, airway, and monitoring by trained providers.

5. POST-CARDIAC ARREST CARE

Optimize organ function, prevent complications, support recovery.

SURVIVAL FACTORS



TIME IS CRITICAL

Survival falls rapidly as no-flow time increases. Begin CPR immediately.



EARLY DEFIBRILLATION

The earliest appropriate shock offers the greatest chance of terminating VF/pVT.



HIGH QUALITY CPR

Depth, rate, full recoil, minimal pauses.



ETCO₂ MONITORING

An abrupt sustained rise may indicate ROSC. Low or falling values should prompt reassessment.



TEAMWORK

Clear roles, closed-loop communication, leadership.



COMPREHENSIVE CARE

Post-ROSC care improves survival and neurologic outcomes.

RHYTHM-BASED ACTION SUMMARY

SHOCKABLE (VF / PULSELESS VT)



- 1 Shock immediately
- 2 CPR for ~2 minutes
- 3 Rhythm check – shock again if still shockable
- 4 CPR + Epinephrine 1 mg IV/IO
- 5 Rhythm check – shock again if still shockable
- 6 CPR + Amiodarone 300 mg IV/IO (or Lidocaine 1–1.5 mg/kg) for refractory VF/pVT
- 7 Continue cycles of CPR, defibrillation, and medications until ROSC

NON-SHOCKABLE (PEA / ASYSTOLE)



- 1 CPR for ~2 minutes
- 2 Epinephrine 1 mg IV/IO every 3–5 minutes
- 3 Consider advanced airway, capnography, and treat reversible causes (Hs & Ts)
- 4 Reassess rhythm every ~2 minutes

CONTINUOUS
HIGH-QUALITY CPR
Reassess rhythm
every ~2 minutes

REVERSIBLE CAUSES (HS & TS)

H – HYPOXIA RELATED



T – OTHER CAUSES



CPR QUALITY CHECKLIST

- ✓ Rate: 100–120 compressions per minute
- ✓ Depth: At least 5 cm (2 in), avoid >6 cm (2.4 in)
- ✓ Allow full chest recoil; do not lean
- ✓ Minimize interruptions (<10 seconds)
- ✓ Ventilation: 30:2 without advanced airway; 1 breath every 6 seconds (10/min) with advanced airway
- ✓ Rotate compressors every 2 minutes
- ✓ Monitor ETCO₂ and pulse oximetry
- ✓ Ensure safety and team communication

DEFIBRILLATION ENERGY (ADULT)

TYPE	ENERGY*
Initial Shock	120–200 J (biphasic)
Second Shock	120–200 J (biphasic)
Subsequent Shocks	120–200 J (biphasic)

*Use the manufacturer's recommended biphasic energy when known; if unknown, use the maximum available setting.

POST-ROSC CARE GOALS

Optimize Oxygenation Target SpO ₂ 90–98% or PaO ₂ 60–105 mmHg. Avoid hyperoxia.	Hemodynamic Support Target MAP ≥65 mmHg. IV fluids and/or vasopressors as needed.	Temperature Control For patients not following commands: 32–37.5°C for at least 36 hours. Prevent fever.	Glucose Control Avoid hypoglycemia and marked hyperglycemia. Manage per protocol.	Treat Cause Identify and treat underlying cause and reversible conditions.	Neurologic Care Provide ongoing assessment and seizure management. Use delayed multimodal prognostication.	Ongoing Care ICU admission, continuous monitoring, rehabilitation, and family communication.
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REMEMBER

High-quality CPR saves lives, but high-quality post-ROSC care saves brains. Every action matters. Every team member matters. Persist with purpose. Follow appropriate termination protocols when indicated.



AVOID COMMON PITFALLS

- ✗ Delaying CPR or defibrillation
- ✗ Inadequate compression depth or rate
- ✗ Excessive interruptions in compressions
- ✗ Failure to treat reversible causes
- ✗ Poor post-ROSC management
- ✗ Failure to recognize ROSC